Note

If A is a non-singular matrix of order 3, then, $|A| \neq 0$. By theorem 1.9 (*ii*), we get $|\text{adj}A| = |A|^2$ and so, $|\text{adj } A|$ is positive. Then, we get $|A| = \pm\sqrt{|\text{adj } A|}$.

So, we get $A^{-1} = \pm\dfrac{1}{\sqrt{|\text{adj } A|}}\text{adj } A$.

Further, by property (*iii*), we get $A = \dfrac{1}{|A|}\text{adj}(\text{adj } A)$.

Hence, if A is a non-singular matrix of order 3, then we get $A = \pm\dfrac{1}{\sqrt{|\text{adj } A|}}\text{adj}(\text{adj } A)$.

Example 1.4

If A is a non-singular matrix of odd order, prove that $|\text{adj } A|$ is positive.

Solution

Let A be a non-singular matrix of order $2m+1$, where $m = 0, 1, 2, \cdots$. Then, we get $|A| \neq 0$ and, by theorem 1.9 (ii), we have $|\text{adj } A| = |A|^{(2m+1)-1} = |A|^{2m}$.

Since $|A|^{2m}$ is always positive, we get that $|\text{adj } A|$ is positive.

Example 1.5

Find a matrix A if $\text{adj}(A) = \begin{bmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{bmatrix}$.

Solution

First, we find $|\text{adj}(A)| = \begin{vmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{vmatrix} = 7(77-35) - 7(-7-77) - 7(-5-121) = 1764 > 0.$

So, we get

$$A = \pm\frac{1}{\sqrt{|\text{adj } A|}}\text{adj}(\text{adj } A) = \pm\frac{1}{\sqrt{1764}}\begin{bmatrix} +(77-35) & -(-7-77) & +(-5-121) \\ -(49+35) & +(49+77) & -(35-77) \\ +(49+77) & -(49-7) & +(77+7) \end{bmatrix}^T$$

$$= \pm\frac{1}{42}\begin{bmatrix} 42 & 84 & -126 \\ -84 & 126 & 42 \\ 126 & -42 & 84 \end{bmatrix}^T = \pm\begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}.$$

Example 1.6

If $\text{adj } A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.

12th_Maths_EM_Vol1_CH 1_Matrices.indd 9 16-12-2022 12:16:02